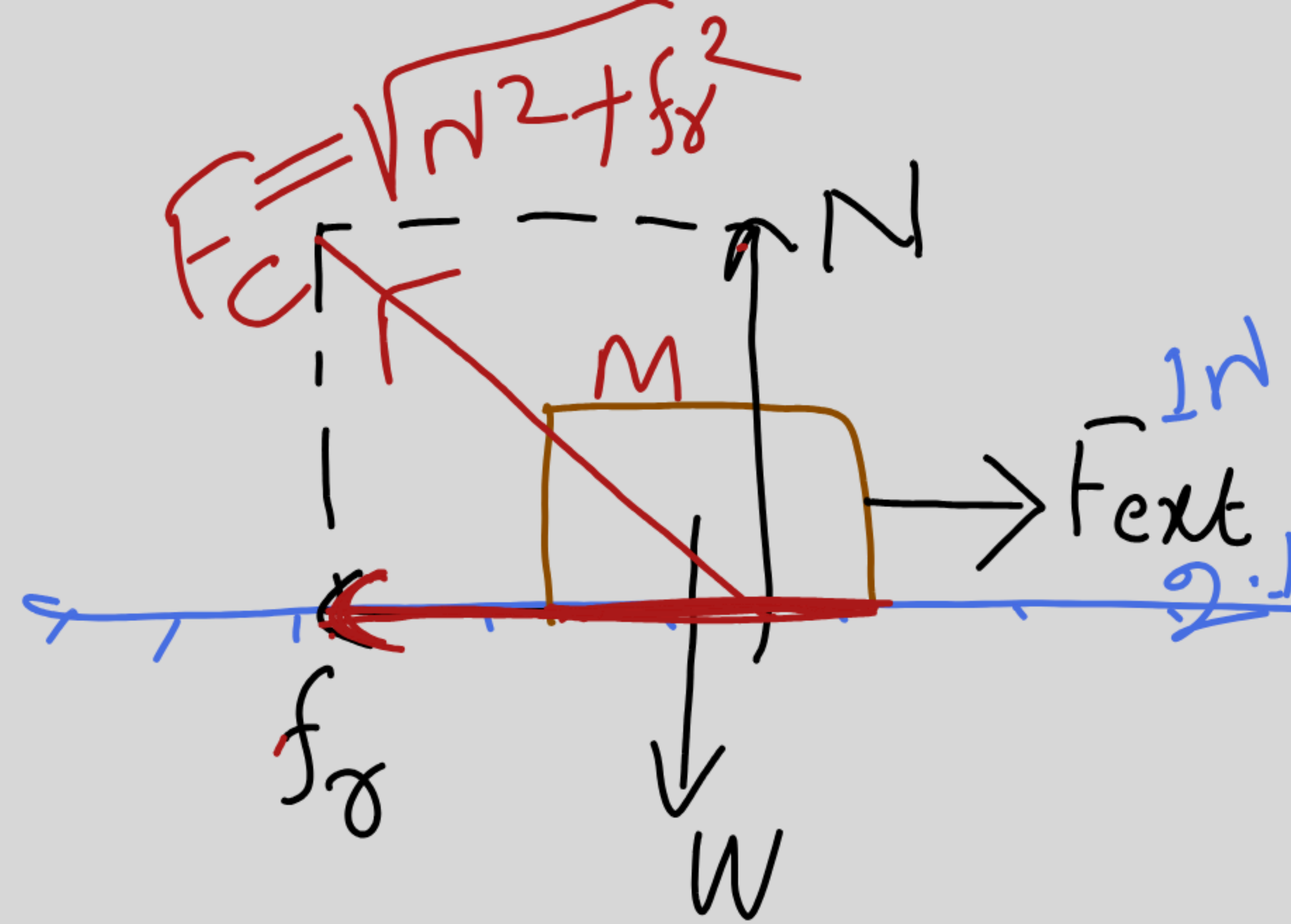




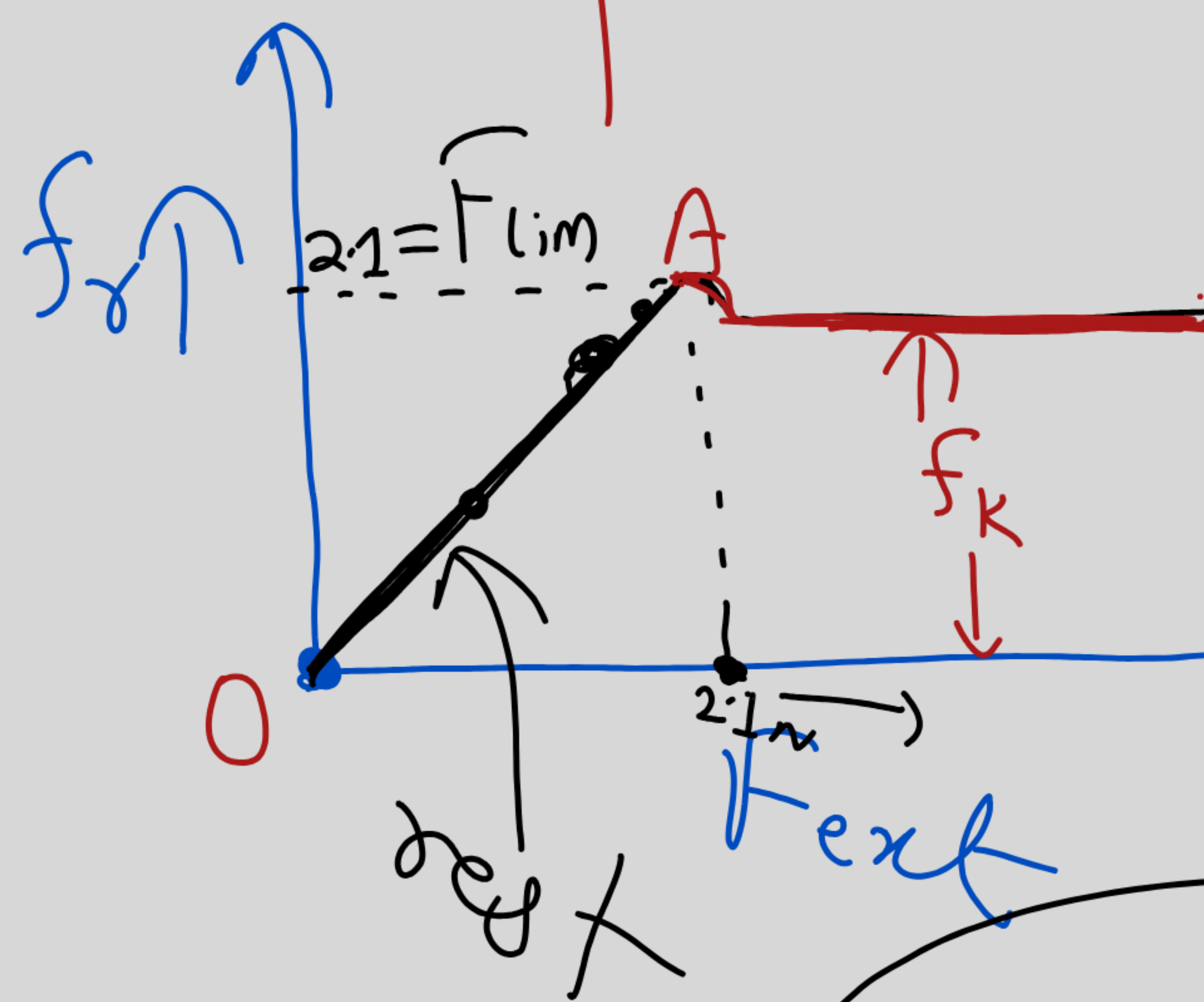
Friction

static friction

F_{ext}	Fric force	cond ⁿ
0	0	rest
1	1	rest
2	2	rest
2.1	2.1	body just tends to move

rest

$f_s \Rightarrow 0 \text{ to } F_{lim}$



2.5	2.005	move
5	2.005	move
10	2.005	move

move
move
move

$$F_{lim} \propto N$$

$$\Rightarrow F_{lim} = \mu_s N$$

$$\Rightarrow \mu_s = \frac{F_{lim}}{N}$$

co-eff of static friction

$$0 \leq f_s \leq F_{lim}$$

$$\mu_s = \mu_k$$

$$f_k \propto N$$

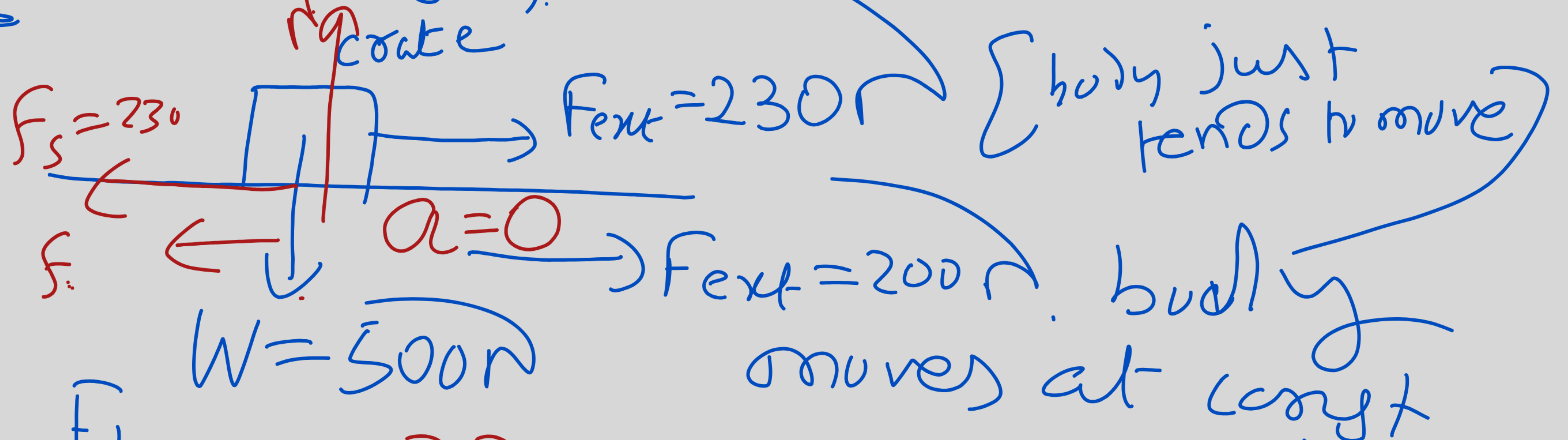
$$f_k = \mu_k N$$

$$\mu_k = \frac{f_k}{N}$$

co-eff of dynamic friction

Q.5 (a) mid sem Exm.

Ex 5.13



$$\mu_s = \frac{F_{lim}}{N} = \frac{230}{500} = 0.46$$

$$\mu_k = \frac{200}{500} = 0.4$$

$v = \text{const.}$ $F_{ext} = 200$
 $a = 0$
 $F_{ext} = ma = 0$

Ex 5.16

$$\alpha = f^n(W \text{ \& } \mu_k)$$

$$N = W \cos \alpha \quad (1)$$

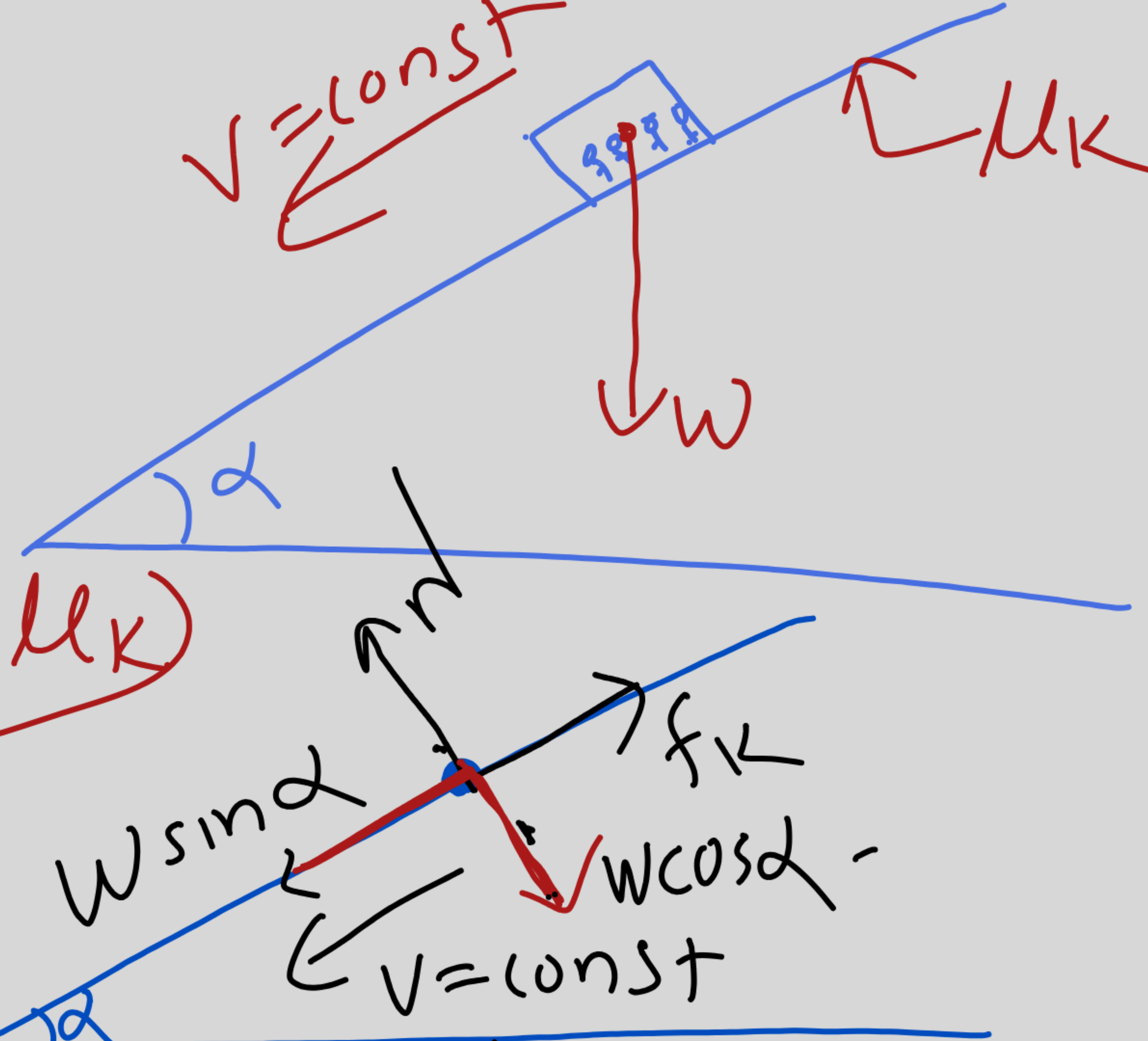
$$f_k = W \sin \alpha$$

$$\Rightarrow \mu_k N = W \sin \alpha$$

$$\Rightarrow \mu_k W \cos \alpha = W \sin \alpha$$

$$\Rightarrow \tan \alpha = \mu_k$$

$$\Rightarrow \alpha = \tan^{-1}(\mu_k)$$



$$a = 0$$

$$F_{\text{ext}} = ma = 0$$

A toboggan loaded with students (total wt W) slides with const vel on a rough surface with const vel having μ_k . Find α in terms of W & μ_k .

$$\mu_k \equiv \frac{f_k}{N}, \quad \mu_s = \frac{F_{\text{lim}}}{N}$$

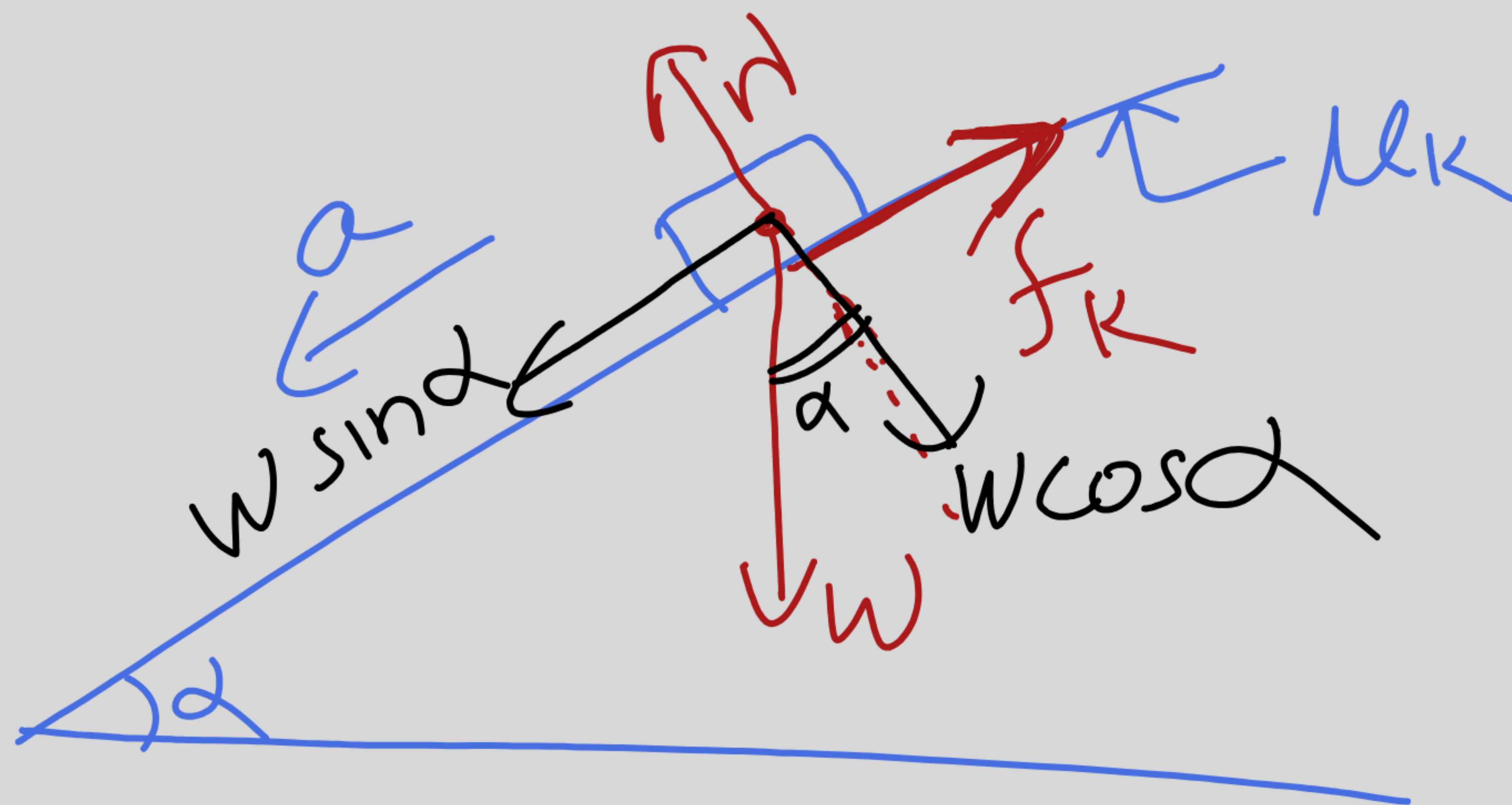
Ex 5.12

$a = ?$

in terms
of g, α, μ_k
& W

$$N = W \cos \alpha \quad (1)$$

$$W \sin \alpha - f_k = \left(\frac{W}{g}\right) a \quad (2)$$

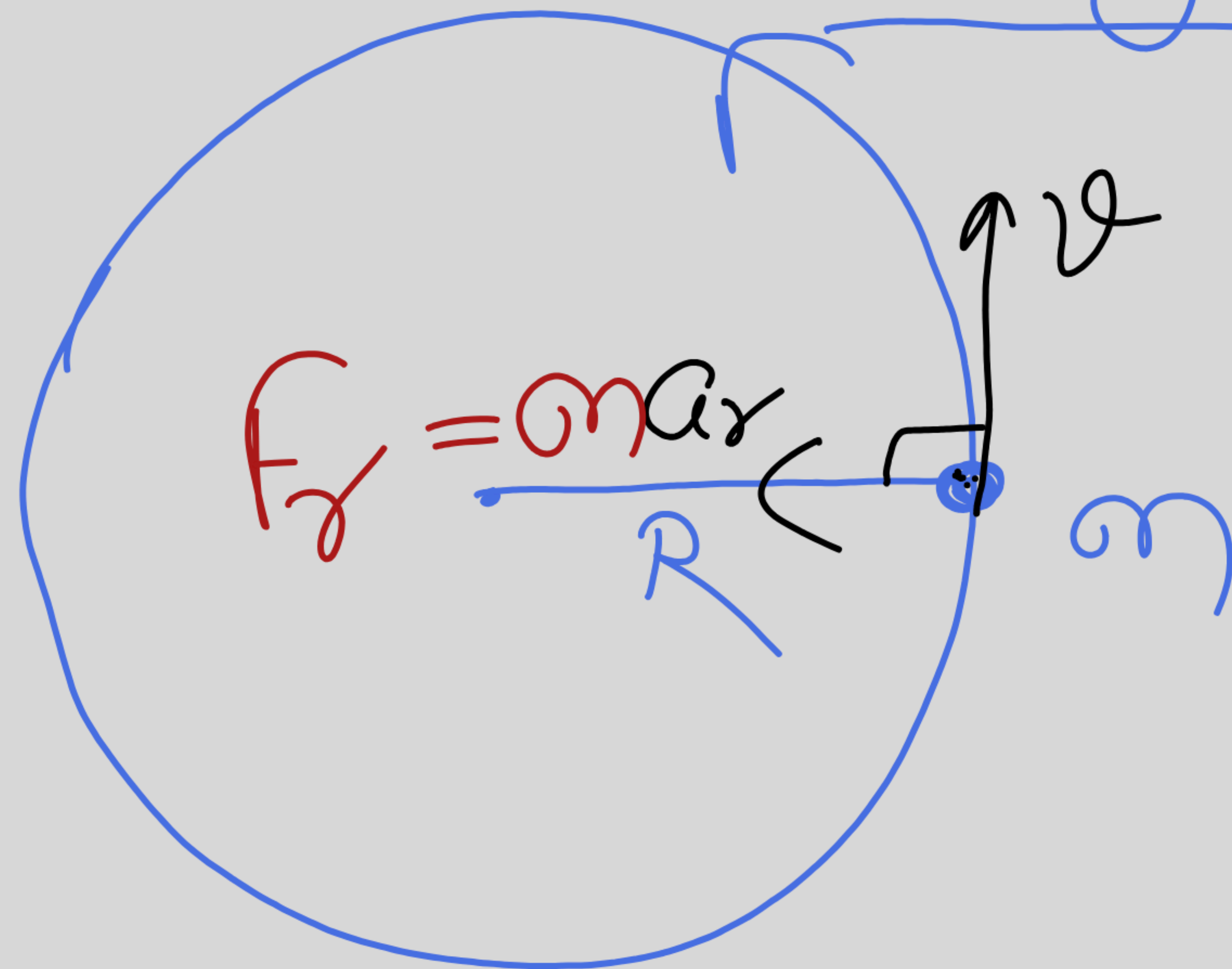


$$\mu_k = \frac{f_k}{N}$$
$$f_k = \mu_k N$$
$$f_k = \mu_k W \cos \alpha$$

$$W \sin \alpha - \mu_k W \cos \alpha = \left(\frac{W}{g}\right) a$$
$$\Rightarrow \sin \alpha - \mu_k \cos \alpha = \left(\frac{W}{g}\right) a$$

$$a = g (\sin \alpha - \mu_k \cos \alpha)$$

Dynamics of CM



$$a_c = \frac{v^2}{R} = \frac{(R\omega)^2}{R} = R\omega^2 = R \frac{4\pi^2}{T^2}$$

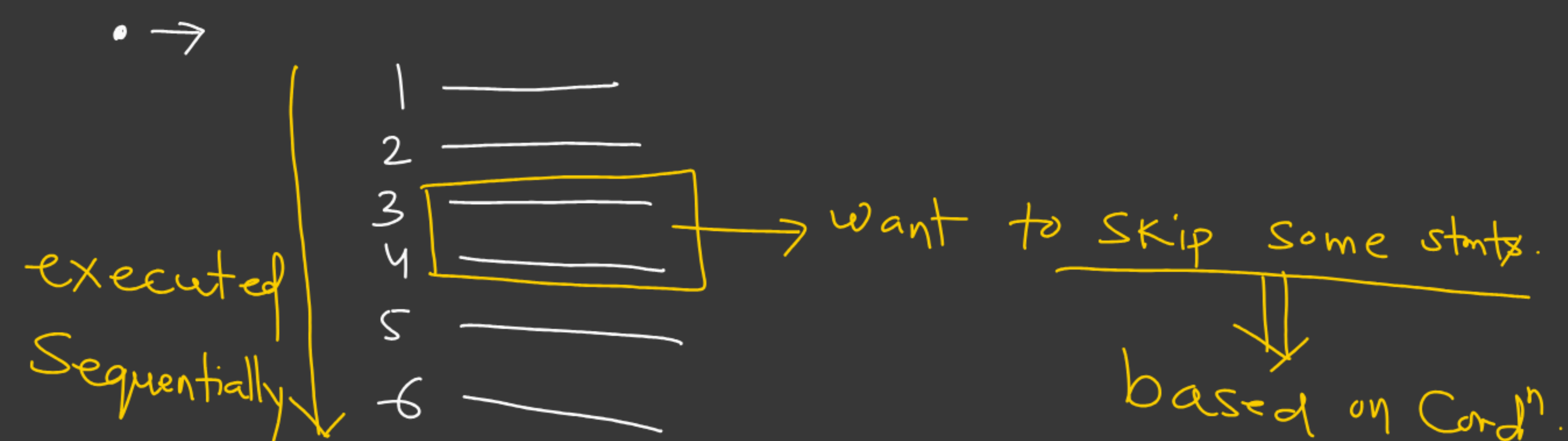
$$\Rightarrow a_c = R 4\pi^2 f^2 \quad \omega = \frac{\theta}{t} = \frac{2\pi}{T}$$

$$F_c = ma_c = m \frac{v^2}{R} = m R \omega^2 = 2\pi f$$

$$= \frac{4\pi^2 m R}{T^2} = 4\pi^2 m R f^2$$

ICP

Conditional statements



Types

- ① Simple if
 - ② if - else
 - ③ if - elseif ladder
 - ④ Switch Case.
- Same Conceptually.

*Keywords

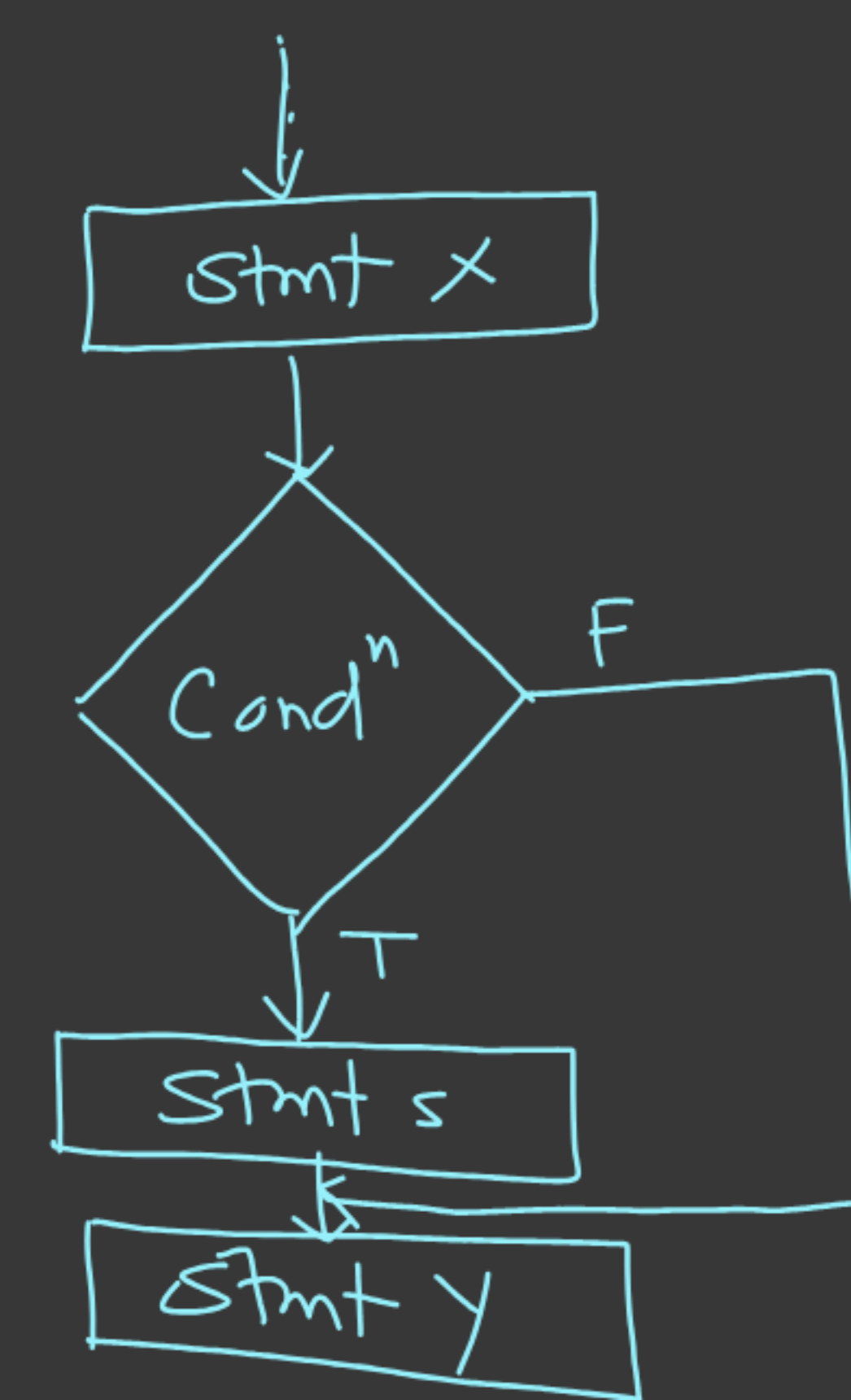
if, else
switch, case
break, default

① Simple if

Syntax

stmt x; → optional
if ($Cond^n$)
{
stmt s; ← stmt/stmts → Compulsory at least one stmt.
}
stmt y → optional

Flow Diagram



```

    ...
    // 38 int temp = sc.nextInt();
    // 38 sc.println("Hello");

    if (temp >= 40)
    {
        // 38 >= 40
        sc.println("Too Hot");
    }
    // 41 sc.println("Thank You");

```

temp = 38
 Hello
 Thank you.

temp = 41
 Hello
 Too Hot
 Thank you.

② if - else.

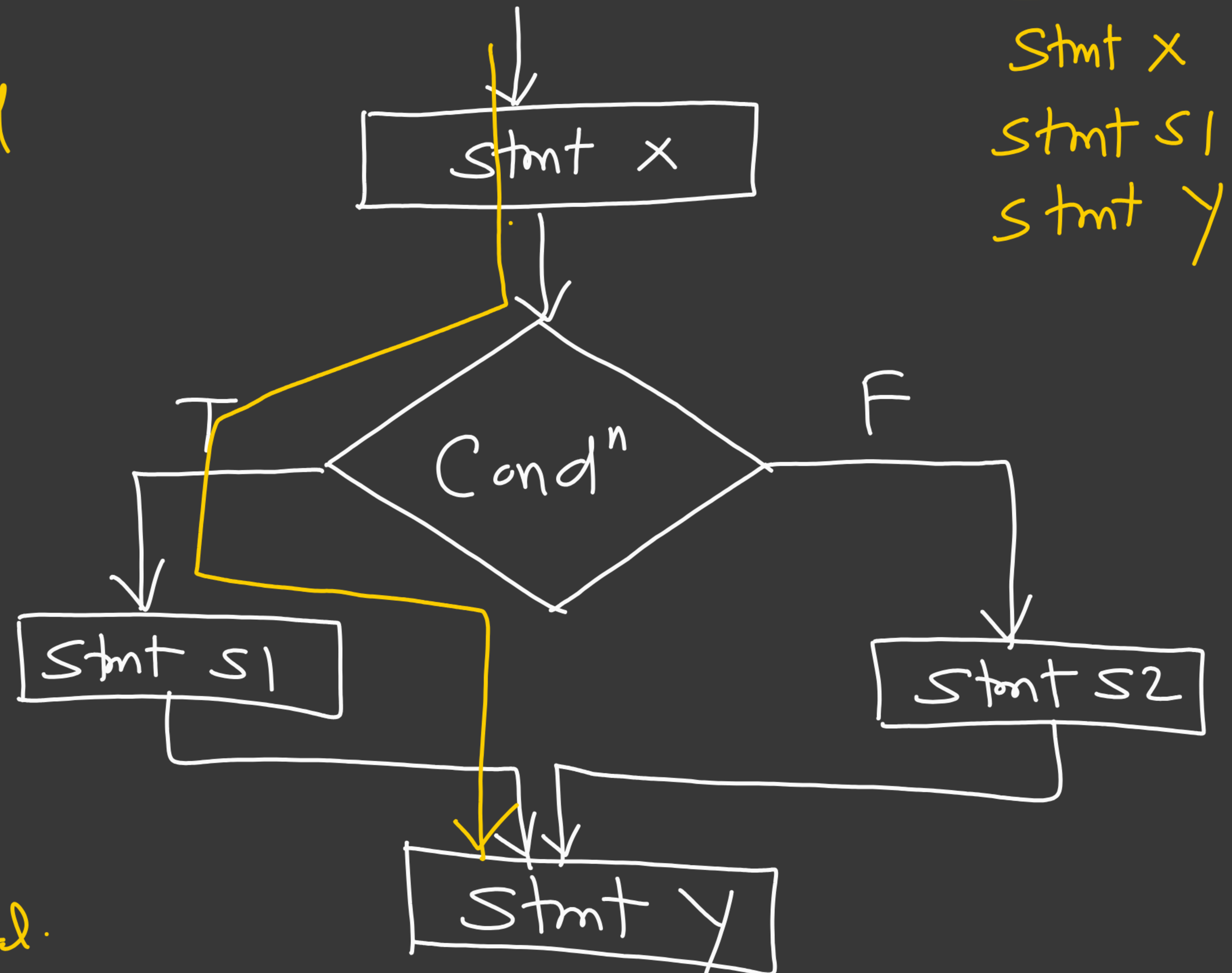
Syntax

```

:
stmt x  $\rightarrow$  optional
if (Condn)
{
    stmt s1
}
else
{
    stmt s2
}
stmt y  $\rightarrow$  optional.

```

Flow Diagram



true
stmt x
stmt s1
stmt y

```

int temp = sc.nextInt();
s.o.pln("Hello");
if (temp >= 35)
{
    41 >= 35
    s.o.pln("Get Cold Drinks");
}
else
{
    s.o.pln("Get Tea");
}
s.o.pln("Thank You");

```

```

temp = 41
Hello
Get Cold Drinks.
Thank You

temp = 16
Hello
Get Tea
Thank You.

```

```

import java.util.Scanner;

public class test
{
    public static void main (String args[])
    {
        Scanner sc = new Scanner(System.in);
        int a, b;
        S.o.pln ("Enter the value of a & b");
        a = sc.nextInt();
        b = sc.nextInt();
        if (a > b)
        {
            S.o.pln ("Greatest Number" + a);
        }
        else
        {
            S.o.pln ("Greatest Number" + b);
        }
    }
}

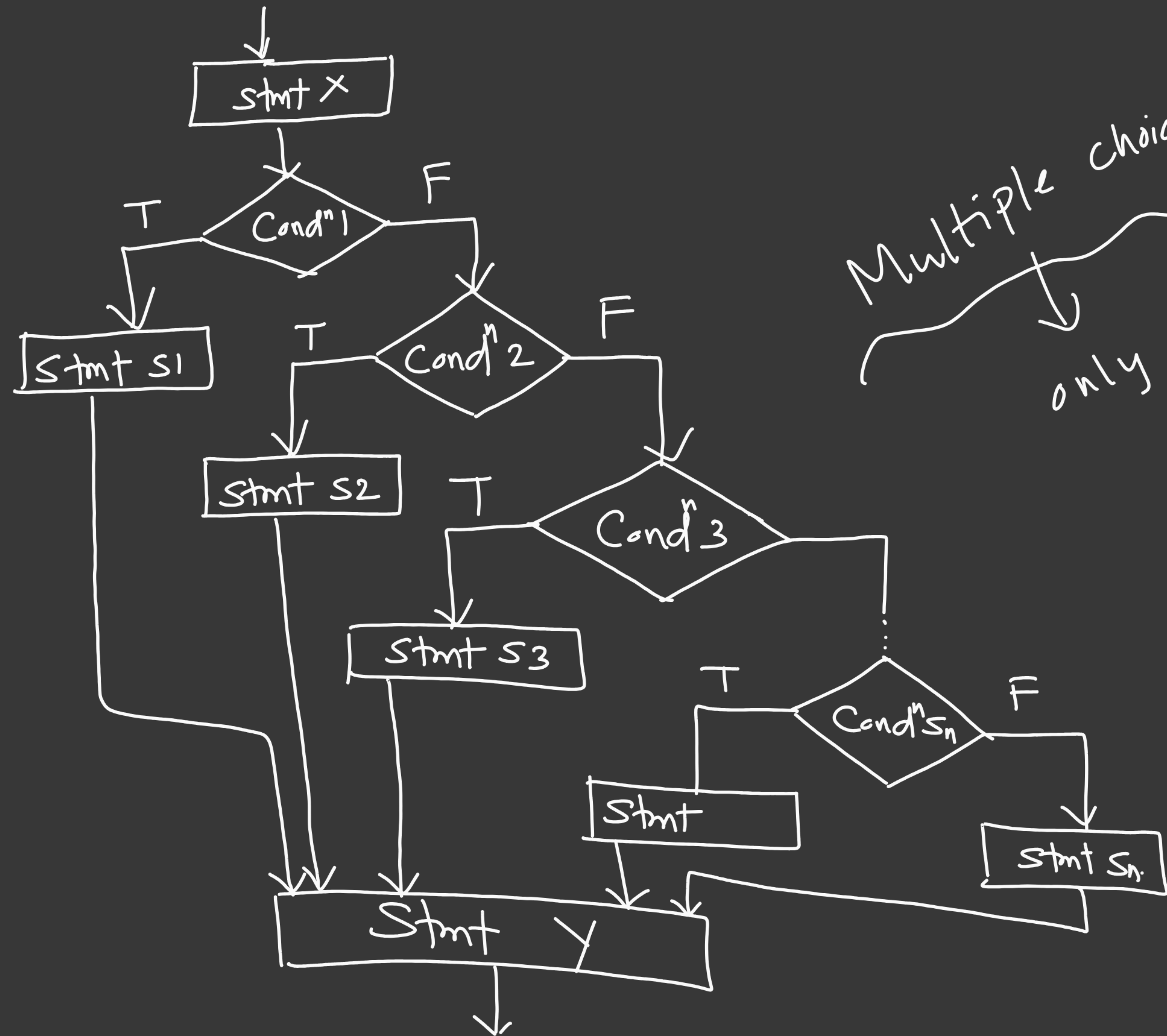
```

- java starts (only Logic)
- java Program.

③ if - else if ladder.

Syntax

```
stmt X  
if (Condn)  
{  
    stmt S1;  
}  
else if (Condn2)  
{  
    stmt S2;  
}  
else if (Condn3)  
{  
    stmt S3;  
}  
...  
else  
{  
    stmt Sn  
}  
stmt Y
```



Multiple choice
↓
only one option.

Day Number

1



Monday

2



Tuesday

w

3



4



Thu

5



Fri

6



Sat

7



Sunday

(Any other
Number)



Out of Range.

```
int dn = sc.nextInt();
```

```
if (dn == 1)
```

```
{
    s.opln("Monday");
}
```

```
(a > b && a > c) ? a : (b > c) ? b : c
```

```
...
else if (n == 7)
{
    s.pln("Sunday");
}
else
{
    s.opn "t of Range";
}
```

Greatest Among 3 no. (a, b, c)

```
if (a > b && a > c)
{
    s.opln("Greatest no" + a);
}
```

```
else if (b > c)
{
    s.opln("Greatest no" + b);
}
```

```
else
{
    s.opln("Greatest no" + c);
}
```


Floor function (Greatest integer function)

$\lfloor x \rfloor =$ Largest integer less than equal to x .

Ceiling function (Least integer function)

$\lceil x \rceil =$ Smallest integer greater than equal to x .

$$\begin{array}{l} \text{Ex } \lfloor 0.3 \rfloor = 0 \quad \lfloor 0.3 \rfloor = 1 \\ \lfloor -1.3 \rfloor = -2 \quad \lfloor -1.3 \rfloor = -1 \\ \lfloor 7.9 \rfloor = 7 \quad \lfloor 7.9 \rfloor = 8 \end{array} \quad \left| \begin{array}{l} \# (x-1) < \lfloor x \rfloor \leq x \\ x \leq \lceil x \rceil < (x+1) \end{array} \right.$$

Prove or disprove

for all real no. x and y .

(a) $\lfloor x+y \rfloor = \lfloor x \rfloor + \lfloor y \rfloor$ (b) $\lceil x+y \rceil = \lceil x \rceil + \lceil y \rceil$

Sol (a) $x=0.5, y=0.5$

$\lfloor x+y \rfloor = \lfloor 0.5+0.5 \rfloor = \lfloor 1 \rfloor = 1$

$\lfloor 0.5 \rfloor + \lfloor 0.5 \rfloor = 0+0=0$

$\lfloor x+y \rfloor \neq \lfloor x \rfloor + \lfloor y \rfloor$

Ex (a) Domain = $\{0, 1, 2, 3, \dots\}$

Range = $\{0, 1, 2, 3, \dots, 8, 9\}$

$f(11) = 1$

$f(100) = 0$

$f(99) = 9$

(b) Domain = $\{1, 2, 3, \dots\}$

Range = $\{2, 3, 4, 5, \dots\}$

$f(00) = 2$

$f(110) = 3$

(c) Domain = $\{0, 01, 00, 11, 101, 1001, \dots\}$

Range = $\{0, 1, 2, 3, \dots\}$

$f(010) = 3$

$f(0) = 1$

(d)

Range = $\{1, 2, 3, \dots\}$

2.5 Sequence and Summation

Sequence

Ex $\{1, 2, 3, 4, \dots\}$ $a_n = n$

$\{1, 3, 5, 7, \dots\}$ $a_n = 2n+1$

$\{2, 4, 6, \dots\}$ $a_n = 2 \cdot n$

$\{1, -1, 1, -1, \dots\}$ $a_n = (-1)^n$
 $n=2, \dots$

$\{1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots\}$ $a_n = \frac{1}{2^n}$

$n=0, 1, 2, \dots$

Ex $1, 2, 2, 3, 3, 3, 4, 4, 4, 4, \dots$?

Sol $1, 2, 2, 3, 3, 3, 4, 4, 4, 4, \dots$

$5, 5, 5, 5, 5, \dots$

